

SATU MARE, Romania

WMO: 150100

Lat: 47.7214N

Lon: 22.8872E

Elev: 124

StdP: 99.84

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.3	-10.8	-16.6	0.9	-13.0	-13.1	1.2	-9.3	10.2	1.6	9.0	1.1	1.4	90	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	33.1	21.5	31.4	21.1	29.9	20.4	23.1	30.0	22.1	28.9	21.2	27.8	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.9	26.5	20.0	14.9	24.9	19.1	14.1	23.7	69.0	30.0	65.3	28.9	62.1	28.0	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.3	DB	-17.5	35.1	5.1	1.8	-21.2	36.3	-24.2	37.4	-27.1	38.4	-30.8	39.7	(4)
(5)			WB	-17.7	24.8	5.1	1.4	-21.3	25.8	-24.3	26.6	-27.1	27.4	-30.8	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.9	1.1	5.5	11.5	16.1	19.7	21.2	21.1	16.2	10.8	5.5	0.4	(6)
(7)		DBStd	8.89	4.90	5.09	4.33	3.84	3.47	3.24	3.03	3.23	3.78	4.19	4.57	4.74	(7)
(8)		HDD10.0	1252	338	249	147	28	2	0	0	0	1	42	146	299	(8)
(9)		HDD18.3	3071	597	482	397	206	86	23	9	11	85	235	384	557	(9)
(10)		CDD10.0	1522	0	0	8	73	191	292	348	344	187	67	12	0	(10)
(11)		CDD18.3	299	0	0	0	1	16	64	98	97	21	1	0	0	(11)
(12)		CDH23.3	3550	0	0	0	27	243	697	1125	1151	289	18	0	0	(12)
(13)		CDH26.7	1310	0	0	0	4	49	227	452	489	88	1	0	0	(13)
(14)	Wind	WSAvg	2.4	2.3	2.5	3.1	2.9	2.7	2.4	2.2	2.1	2.2	2.1	2.3	2.2	(14)
(15)	Precipitation	PrecAvg	594	38	31	35	43	65	83	65	64	37	38	47	49	(15)
(16)		PrecMax	842	82	73	118	108	127	221	184	127	129	155	109	102	(16)
(17)		PrecMin	378	3	0	1	6	25	20	17	6	0	1	9	1	(17)
(18)		PrecStd	108	20	17	26	20	30	47	36	31	27	30	29	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	14.7	20.0	26.6	30.0	33.1	35.2	35.9	32.3	25.2	19.1	12.2	(19)
(20)			MCWB	8.5	9.7	11.9	16.4	19.0	22.2	22.0	21.2	19.6	16.4	13.0	9.5	(20)
(21)		2%	DB	9.0	11.8	17.2	23.2	28.0	31.1	32.9	33.6	29.0	23.1	16.8	10.1	(21)
(22)			MCWB	7.3	8.4	10.8	14.4	18.0	21.7	21.9	21.3	18.1	15.7	12.4	7.9	(22)
(23)		5%	DB	7.0	9.6	14.9	21.3	26.1	29.2	31.1	31.8	26.9	20.9	14.2	8.1	(23)
(24)			MCWB	5.9	7.3	9.5	13.4	17.4	20.5	21.0	21.1	17.7	15.1	11.1	6.9	(24)
(25)		10%	DB	5.2	7.8	12.8	19.3	24.1	27.4	29.2	29.4	24.1	18.1	12.2	6.2	(25)
(26)			MCWB	4.2	6.2	8.6	12.4	16.8	19.8	20.4	20.3	16.9	13.7	9.8	5.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	10.4	13.1	17.1	20.6	24.1	24.3	24.3	20.5	17.9	14.4	9.9	(27)
(28)			MCD B	10.3	13.3	17.9	24.7	26.8	30.3	30.7	30.9	28.3	22.4	17.4	11.3	(28)
(29)		2%	WB	7.4	8.7	11.3	15.5	19.3	22.7	23.0	23.0	19.4	16.5	12.9	8.1	(29)
(30)			MCD B	8.5	11.2	15.7	21.9	25.6	28.9	29.8	30.4	26.4	21.3	15.8	9.5	(30)
(31)		5%	WB	5.8	7.4	10.2	14.1	18.2	21.6	22.0	21.8	18.5	15.5	11.5	6.8	(31)
(32)			MCD B	6.7	9.2	14.0	19.7	24.3	27.4	29.3	28.5	24.7	19.8	14.1	7.9	(32)
(33)		10%	WB	4.3	6.1	9.0	13.1	17.3	20.5	21.1	20.9	17.5	14.3	10.2	5.4	(33)
(34)			MCD B	5.1	7.4	12.1	18.1	22.9	26.0	27.8	27.9	22.9	17.8	12.1	6.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.6	9.9	11.9	12.2	12.1	12.9	13.7	12.3	11.3	8.5	5.7	(35)
(36)			MCDBR	7.7	10.1	13.8	15.2	15.8	15.2	16.1	16.8	16.6	15.2	11.4	7.7	(36)
(37)			MCWBR	5.9	7.1	8.1	7.5	7.3	6.6	6.1	6.0	6.8	8.4	7.4	5.9	(37)
(38)		5% WB	MCDBR	7.2	9.4	12.2	13.7	13.9	13.4	14.5	14.5	13.9	13.0	10.0	7.1	(38)
(39)			MCWBR	5.7	6.8	7.6	7.1	7.0	6.4	6.1	6.4	6.4	7.5	6.6	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.355	0.394	0.445	0.429	0.429	0.449	0.448	0.411	0.382	0.356	0.329	(40)	
(41)		taud	2.289	2.251	2.204	2.097	2.209	2.239	2.196	2.194	2.262	2.314	2.330	2.328	(41)	
(42)		Ebn at Noon	726	789	813	806	837	837	813	795	789	749	689	678	(42)	
(43)		Edh at Noon	77	99	121	148	137	134	138	133	113	92	74	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.01	1.83	3.05	4.42	5.41	5.89	5.78	5.31	3.69	2.35	1.28	0.75	(44)	
(45)		RadStd	0.16	0.28	0.45	0.47	0.56	0.39	0.52	0.48	0.46	0.31	0.18	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.87	N/A	N/A	+1.31	N/A	N/A	-126	-206	+136	+68	(46)
(47)	Regional (48 neighbors)	+0.69	N/A	N/A	+1.11	+0.47	N/A	-137	-205	+114	+53	(47)

Nomenclature: See separate page